

Introduction to Mathematics and Optimization

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The references in the suggested answers to this exam set refer to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (6.12.2023) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2$$

and let $C \subseteq \mathbb{R}^3$ be the subset

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid y + z \geq 1, x + z \geq 1, 2x + y + z \leq -1, x + 2y + z \leq -1\}.$$

We consider the optimization problem

$$\begin{aligned} \min f(x, y, z) \\ (x, y, z) \in C. \end{aligned} \tag{1}$$

(a) Show that f is a convex function, but not a strictly convex function.

ANSWER:

The Hessian matrix of f is

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This is a diagonal matrix with nonnegative entries and therefore positive semidefinite according to Exercise 8.26. Hence f is a convex function by Theorem 8.21. f cannot be a strictly convex function, since $f(0, 0, z) = 0$ for all $z \in \mathbb{R}$. The conditions in Definition 6.1 are not satisfied. Here one can, for example, choose $x = (0, 0, 0)$ and $y = (0, 0, 1)$ as well as $t = 1/2$.

(b) Use Fourier-Motzkin elimination of z to show that $(x, y, z) \in C$ implies $x \leq -1$ and $y \leq -1$. Can one conclude from this that C is not bounded?

ANSWER:

The inequalities are

$$\begin{aligned} y + z &\geq 1 \\ x + z &\geq 1 \\ 2x + y + z &\leq -1 \\ x + 2y + z &\leq -1 \end{aligned}$$

We obtain

$$\begin{aligned} 1 - y &\leq z \\ 1 - x &\leq z \\ z &\leq -1 - 2x - y \\ z &\leq -1 - x - 2y \end{aligned}$$

Elimination of z gives the inequalities (according to Section 4.5)

$$\begin{aligned}x &\leq -1 \\x + y &\leq -2 \\y &\leq -1\end{aligned}$$

Here the inequality $x + y \leq -2$ is a consequence of the other two inequalities. Thus C is not bounded, since for each $x \leq -1$ and $y \leq -1$ we can find a z such that $(x, y, z) \in C$. See Section 5.2.7. For example, for $v = (-n, -n, z)$ we have $|v| = \sqrt{2n^2 + z^2} \geq n$ for all $n \geq 1$. Thus $|v|$ is not bounded for $n \geq 1$. Note that we have chosen z so that $v \in C$.

(c) Show that (1) is a convex optimization problem.

ANSWER:

This is a standard exercise. See the answers to many other similar old exam problems.

(d) Write down the KKT conditions for the optimization problem (1). Use only references to and concepts from the syllabus.

ANSWER:

In accordance with (9.24) we set up the KKT conditions for the optimization problem:

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ \lambda_3 &\geq 0 \\ \lambda_4 &\geq 0 \\ -y - z + 1 &\leq 0 \\ -x - z + 1 &\leq 0 \\ 2x + y + z + 1 &\leq 0 \\ x + 2y + z + 1 &\leq 0 \\ \lambda_1(-y - z + 1) &= 0 \\ \lambda_2(-x - z + 1) &= 0 \\ \lambda_3(2x + y + z + 1) &= 0 \\ \lambda_4(x + 2y + z + 1) &= 0 \\ 2x - \lambda_2 + 2\lambda_3 + \lambda_4 &= 0 \\ 2y - \lambda_1 + \lambda_3 + 2\lambda_4 &= 0 \\ -\lambda_1 - \lambda_2 + \lambda_3 + \lambda_4 &= 0\end{aligned}$$

(e) Use the KKT conditions to show that $(-2, -2, 4)$ is not a solution to the optimization problem (1).

ANSWER:

By substituting $x = -2, y = -2$ and $z = 4$ into the KKT conditions we obtain

$$\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0.$$

The penultimate equation in the KKT conditions yields

$$-4 = 0,$$

which is a contradiction. Thus $(-2, -2, 4)$ is not a solution to the optimization problem (1) with reference to Theorem 9.30(i). We must check that the optimization problem is strictly feasible to be able to apply Theorem 9.30(i). Here we can in fact choose $x = -2, y = -2$ and $z = 4$, which yields an interior point in C according to Definition 9.29.

- (f) It is known that $(-1, -1, 2)$ is a solution to the optimization problem (1). Are there more than one solution to the KKT conditions for this optimal solution?

ANSWER:

By substituting $x = -1, y = -1$ and $z = 2$ into the KKT conditions we obtain in the last three equations

$$\begin{aligned} -\lambda_2 + 2\lambda_3 + \lambda_4 &= 2, \\ -\lambda_1 + \lambda_3 + 2\lambda_4 &= 2, \\ -\lambda_1 - \lambda_2 + \lambda_3 + \lambda_4 &= 0. \end{aligned}$$

These equations are satisfied for

$$\begin{aligned} \lambda_2 &= 2 - \lambda_1, \\ \lambda_3 &= 2 - \lambda_1, \\ \lambda_4 &= \lambda_1. \end{aligned}$$

Thus there are infinitely many solutions to the KKT conditions for the optimal solution $(-1, -1, 2)$ corresponding to $\lambda_1 \in [0, 2]$.

PROBLEM 2

For $a \in \mathbb{R}$ the matrix M_a is defined by

$$M_a = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{pmatrix}.$$

For example

$$M_0 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

- (a) Use an algorithm from the syllabus to show that M_1 is positive semidefinite.

ANSWER:

We diagonalize M_a in full generality by using the algorithm in Section 8.6 of the book. We transform M_a by first subtracting the first row from the two other rows and then subtracting the first column from the two other columns. This yields the diagonal matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & a-1 \end{pmatrix}.$$

In line with the answers to many other old exam problems we then know that M_a is positive semidefinite if and only if $a \geq 1$ and positive definite if and only if $a > 1$. Thus M_1 is positive semidefinite. Here it is also important to refer to Proposition 3.41.

- (b) Find the set of all $a \in \mathbb{R}$ such that M_a is positive definite, again using the algorithm from the syllabus.

ANSWER:

This set is $(1, \infty)$ as a continuation of the answer to part (a).

- (c) Is M_1 invertible?

ANSWER:

M_1 is not invertible, since as a continuation of the answer to part (a) we know that there exists an invertible matrix B such that

$$B^T M_1 B = C := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Here the matrix C is not invertible, since for instance $C \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$. If M_1 were invertible, then $B^T M_1 B$ would also be invertible, since a product of invertible matrices is invertible.

- (d) Find a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that M_2 is the Hessian matrix of f . Is this function strictly convex?

ANSWER:

Such a function could be

$$f(x, y, z) = \frac{1}{2}x^2 + y^2 + z^2 + xy + xz + yz$$

according to Definition 8.2. This function is strictly convex according to Theorem 8.21, since M_2 is positive definite following the answer to part (a).

PROBLEM 3

Consider the function $f_a : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$f_a(x, y) = x^2 + axy + y^2$$

where $a \in \mathbb{R}$ is a real constant. For example $f_2(x, y) = x^2 + 2xy + y^2$. Also let

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 0, y \leq 0, x + y \geq -2\}.$$

(a) Show that C is bounded using the definition in the syllabus.

ANSWER:

Using Fourier-Motzkin elimination (Section 4.5) we obtain the inequalities

$$\begin{aligned}x &\leq 0 \\y &\leq 0 \\y &\geq -2 - x\end{aligned}$$

Eliminating y gives the inequalities

$$\begin{aligned}x &\leq 0 \\-2 - x &\leq 0\end{aligned}$$

i.e. $-2 \leq x \leq 0$. Eliminating x gives the inequalities

$$\begin{aligned}y &\leq 0 \\y &\geq -2\end{aligned}$$

i.e. $-2 \leq y \leq 0$. Thus C is bounded with $M = 4$ in Equation 5.16.

(b) Investigate for which values of a the function f_a is convex.

ANSWER:

The Hessian matrix of f is

$$\begin{pmatrix} 2 & a \\ a & 2 \end{pmatrix}$$

This can, using the algorithm in Section 8.6 of the book, be transformed into

$$\begin{pmatrix} 2 & a \\ 0 & 2 - \frac{a^2}{2} \end{pmatrix},$$

which is positive semidefinite if and only if $2 \geq 0$ and $2 - \frac{a^2}{2} \geq 0$. Thus f_a is convex if and only if $a^2 \leq 4$ and hence $a \in [-2, 2]$. An important detail is that one cannot use Exercise 3.42 here. One has to rely on Proposition 3.41 and Exercise 8.26.

(c) Find all solutions to the optimization problem

$$\begin{aligned}\min & f_{-2}(x, y) \\ & (x, y) \in C.\end{aligned}$$

ANSWER:

We have $f_{-2}(x, y) = x^2 - 2xy + y^2 = (x - y)^2$. Thus all solutions to the optimization problem are given by $\{(x, x) \mid x \in \mathbb{R}\} \cap C = \{(x, x) \mid x \in [-1, 0]\}$.